GLFRNet: Global–Local Feature Refusion Network for Remote Sensing Image Instance Segmentation

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***Abstract*— Instance segmentation is a significant way for remote sensing image (RSI) interpretation. The large number, sharp variation of sizes, and complex background of objects raise higher demands for instance segmentation models. The synergistic usage of global and local features has drawn great attention due to its superior performance but has not been fully explored in mainstream instance segmentation methods. In this work, a global–local feature refusion network (GLFRNet) with two fusion procedures is proposed to fully utilize coarse-grained and fine-grained features for RSI instance segmentation. In this model, the backbone integrates both convolutional neural network (CNN)-based and VMamba-based branches to extract local and global features, respectively. Three novel models are proposed to leverage the features adaptively, i.e., the cross-dim feature fusion (CDFF) module, the semantic complementary feature fusion (SCFF) module, and the guided feature refusion module (GFRM). The CDFF module is designed to aggregate features flexibly by fusing features from two backbones with different attention modules in the first fusion procedure. The GFRM and SCFF module are proposed in the refusion procedure to generate accurate segmentation results. Inspired by agent attention, the GFRM dynamically assembles detailed features for mask generation by refusing local and global features with the guidance of fusion results from CDFF. The SCFF module complements the significant features by enhancing and integrating global, local, and detailed features, and finally generates masks of instances. Extensive experiments demonstrate that GLFRNet outperforms the second-best model by 1.9, 1.3, and 0.3 in mask average precisions (APs) on NWPU VHR-10, WHU Building, and iSAID datasets.**

***Index Terms*— Agent attention, instance segmentation, remote sensing images (RSIs), VMamba.**

I. Introduction

THE advancement of the quality of remote sensing images (RSIs) has broadened the scenes of their applications. With different requirements, various approaches in computer vision are used to conduct auto interpretation of RSI, such as semantic segmentation [1], [2], [4] and object detection [5], [6], [7]. With the ability to achieve all of the localization, classification, and pixel-level segmentation of instances, instance segmentation approaches have been widely applied to interpret RSI images [8], [9]. Many approaches to instance segmentation of natural images have been proposed. The two-stage and one-stage frameworks are two main categories of them. Methods based on two-stage framework [10], [11], [12], [13] first generate proposals from feature maps and then conduct regression and segmentation based on the regions of interest (RoIs) from proposals. Feature pyramid network (FPN) [14] is usually applied to improve the performances. The one-stage framework [15], [16], [17], [18] generates masks without proposals and achieves comparative results to the two-stage methods while saving computing resources.

However, the performance of these methods decreases when applied to RSIs. Several issues specific to RSIs are not fully considered in these methods, i.e., the significant variation in scales and shapes of objects, the dense distribution of objects, and the complex background. Many studies have been conducted to overcome these issues from perspectives of both enhancing multiscale perception ability and augmenting detailed recognition ability to details of detection and segmentation modules. These methods usually improve FPN to enhance the ability of multiscale perception and modify regression and segmentation heads to augment the ability of detailed perceptual [8], [9], [19], [20], [21], [22], [23], [24], [25]. Experiments demonstrate the effectiveness of these strategies. Nevertheless, these improvements are based on a single backbone, and the performance improvement is limited.

The application of different backbones in the single-branch architecture could affect the performances of detection and segmentation significantly since the backbone is the sole source of image features. Generally, the local features could be obtained from backbones based on convolutional neural network (CNN) such as ResNet [26]. However, such a scheme lacks semantic modeling in the global view, which could cause accuracy loss in localization and segmentation. On the other hand, the global context features could be generated from Transformer-based backbones such as ViT [27], but its application could not satisfy the need for detailed local features. To accommodate information requirements in both global and local aspects, the integration of both types of backbones could be the solution.

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Many studies have been conducted on the collaborative usage of multiple backbones for semantic segmentation [1], [2], [3], [4] and object detection [5], [6]. However, these methods could not be simply applied to the instance segmentation task, because of the omission of required features, either instance-level or pixel-level. Besides, simple application of pixel addition or concatenation is insufficient, which could not keep enough informative features, and lead to confusion and suppression of global and local features.

In this article, the global–local feature refusion network (GLFRNet) is proposed to achieve an efficient application for global and local features in RSI instance segmentation. The network is based on the two-stage framework. The backbone includes two branches to extract global and local features. VMamba-t [28] is applied in the global branch to achieve efficient feature extraction with lower computational cost, and ResNet50 [26] is used in the local branch. With the two types of features, the regression and segmentation tasks are achieved through the refusion pipeline. In the pipeline, proposals are generated according to the enriched features from the first fusion of two branches of backbones, and masks are generated from the other fusion procedure to reserve more detailed information. Three new modules, which are the cross-dim feature fusion (CDFF) module, semantic complementary feature fusion (SCFF) module, and guided feature refusion module (GFRM), are introduced in two fusion procedures. The CDFF module in the first procedure achieves a fusion of global and local features from two backbones to gather significant features for flexible proposal generation. The refusion procedure for mask generation consists of SCFF module and GFRM. Inspired by the agent attention [29], the GFRM dynamically selects and assembles detailed features from global and local RoI features with the guidance of fused features. The SCFF module complements the necessary features from global, local, and detailed features to generate the mask. It conducts a self-enhancement procedure on global and local features, and then combines them and the output of GFRM to complement necessary features and generate the segmentation results. Comparative experiments are conducted on iSAID dataset [30], NWPU VHR-10 dataset [31], [32], and WHU Building dataset [33], demonstrating that our model surpasses existing state-of-the-art (SOTA) methods, and multiple ablation experiments are performed to validate the effectiveness of our approach.

The main contributions are summarized as follows.

1) An RSI instance segmentation framework, GLFRNet, is proposed. Two fusion procedures are integrated in this model, to achieve efficient feature fusion of global and local feature maps, and recapture the detailed information for mask generation.
2) In the first fusion procedure, the CDFF module is proposed to gather significant features from global and local feature maps from backbones and generate more informative features for proposal generation.
3) In the refusion procedure, the global feature refusion module and SCFF module are proposed to gather the detailed content from feature maps and complement the necessary features of global, local, and detail to generate masks.
4) Comparative experiments on three datasets show that our method surpasses existing SOTA methods.

The structure of this article is organized as follows. In Section II, some relevant literature is reviewed. In Section III, the proposed method will be thoroughly introduced, including the overall architecture and three new modules. In Section IV, comparative and ablation experiments are reported. In the end, the conclusion is summarized in Section V.

II. Related Work

A. Applications of Dual-Branch Backbone

Besides improving the modules in the model, it has been gaining more attention to expand ways to obtain various kinds of features from the input images. A common approach is to apply dual backbones in the model framework, which has been adopted by many models for related tasks. For object detection of RSI, ASNet [5] used pyramid vision transformer (PVT) and ResNet18 to achieve salient object detection, and DMFNet [6] used two encoders to improve infrared small target detection. For semantic segmentation, FTransUNet [1] used two ResNet backbones to extract visible image features and digital surface model (DSM) features to improve the performance of semantic segmentation, DPFANet [4] combined CNN and Swin Transformer to extract both global and local features, and DFF-Net [3] applied ResNet34 and ShuffleNet to achieve small object segmentation. Such design had also been applied in instance segmentation models in some fields. HDA-Net [34] used a dual-branch backbone to finish nuclei instance segmentation, and CalibNet [35] used two backbones to process RGB and depth images. These studies all demonstrate the superiority of combining different features. However, this design has not been fully studied in RSI instance segmentation. These methods could not be applied to this task simply. For the methods not specified for instance segmentation, they neglect the coordination of detection and segmentation. The methods for instance segmentation in other fields, on the other hand, lack consideration of the difficulties in RSI processing, such as dense distribution of instances and complex background. Thus, the application of the dual-branch backbone in remote sensing instance segmentation is still to be studied.

B. Visual Applications of Mamba

Mamba [36] is a new type of state-space model for natural language processing. With the applications of selective mechanism and hardware-aware algorithm, it achieves efficient modeling for long-term dependencies with linear complexity. Inspired by the success of language modeling, multiple studies have been conducted on transferring Mamba into the computer vision field. VMamba [28] proposed the 2-D selective scan to balance the computational cost and the requirement for global context interaction. First, the input image is split into patches, and then, the compressed hidden states are calculated for each patch along four specific paths to get the results.

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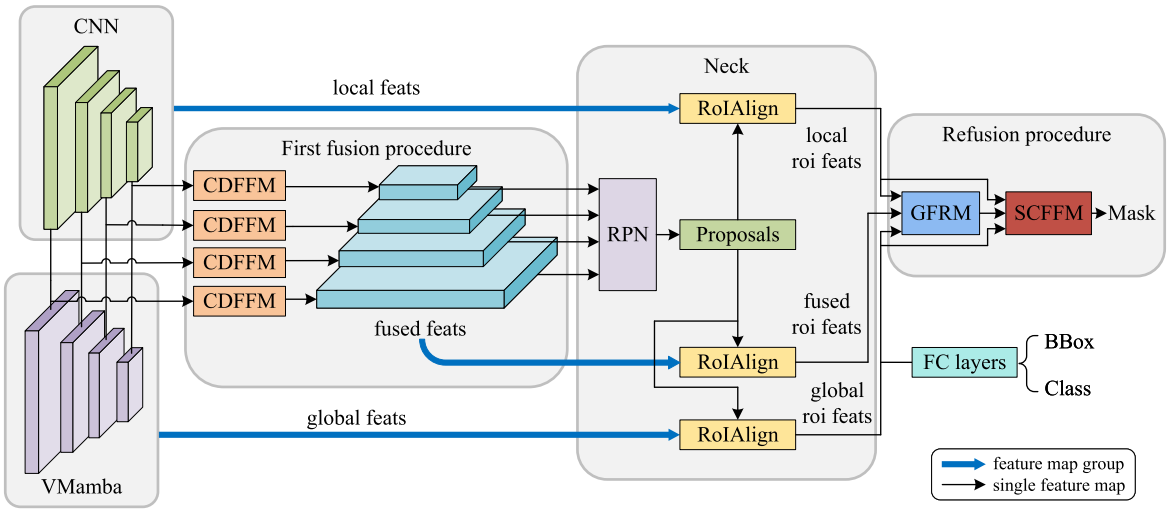


Fig. 1. Overall architecture of GLFRNet. The input image is processed by the two branches of backbone, to generate local and global features. The CDFF module conducts the first fusion procedure of two types of features and generates fused features. The RPN is conducted on the fused features. RoIAlign is applied to the three groups of features to generate global, local, and fused RoI features. The GFRM and SCFF module finally generate mask prediction.

Vision Mamba [37] also applied a patchifying strategy and bidirectional scanning. It also used positional embedding to enhance spatial information. LocalMamba [38] split the input image into distinct windows and scanned each of them individually, to enhance local detailed extraction. It also proposed a method to search efficient scanning directions. EfficientVMamba [39] used a skip-sampling mechanism to save computational cost and integrated the convolution branch into the network. And SE module was applied to fuse the results. These studies demonstrate the advantages of VMamba in global feature modeling and efficiency. In this model, we use VMamba as the backbone to leverage the advantages.

C. Instance Segmentation in Remote Sensing

To process objects with dense distribution, large variations in scales, and complex backgrounds, methods specified on RSI instance segmentation need the higher ability of multiscale perception and fine-grained feature distinction. Most methods are improved based on frameworks with FPN. FB-ISNet [24] uses BiFPN to enhance multiscale feature fusion and optimizes the detection and segmentation heads to alleviate the parameters. Chen et al. [21] propose to fuse the design of Mask R-CNN and MPViT to improve multiscale feature representation ability. LFG-Net [23] proposes LFCP to extend FPN features to a lower level and uses HR-FIM to guide the segmentation task. DB-BlendMask [25] proposes SBC to adjust weight allocation for the targets and introduces CBM to improve semantic representation with classification results. ISDNet [20] applies dilated convolutions and residual connections to HTC to improve the performance. RSIISN [8] proposes CAS to filter significant features after FPN, MSA to enhance multiscale feature fusion, and SRL to fuse the detailed features and global relationship features to generate masks. Liu et al. [40] apply the adaptive feature extraction network, guided anchoring module, and iterative subdivision mask module to finish the task. CATNet [22] proposes DenseFPN, SCP, and HRoIE to improve the performance of the model.

These methods had explored many approaches for efficient fusion of multiscale features and segmentation accuracy improvement. However, these methods were based on a single backbone and did not consider the application of multiple backbones. With features from different types of backbones, it is still to be studied how to gather necessary features efficiently and improve segmentation accuracy. In this article, GLFRNet is proposed to solve these problems.

III. Method

In this section, a detailed exposition of GLFRNet will be provided. In Section III-A, we will introduce the overall architecture of the model. Then, the design of the CDFF module and the SCFF mask head will be described in Sections III-B and III-C, respectively. In Section III-D, the GFR procedure will be introduced.

A. Overall Architecture

The overall architecture of the model is shown in Fig. 1. The model is based on the two-stage framework. The backbone with two branches first extracts features from the input image and then generates global and local feature map sets. Then, with two types of features, the CDFF module is applied to conduct the first fusion procedure at each level. The results from the CDFF module, namely, fused features, are used in the region proposal network (RPN) module to generate proposals. The proposals are applied to three groups of feature maps for RoI feature extraction, respectively, to generate global, local, and fused RoI features. Finally, the refusion procedure with SCFF module and GFRM is conducted on three groups of RoI features to predict high-quality masks. Regression and classification are finished only on global RoI features.

In the backbone, two branches are used for local and global feature extractions, respectively. The local branch uses ResNet50 to preserve the details of images. For the global branch, Transformer has been proven superior at global feature extraction, but the high computational cost blocks its

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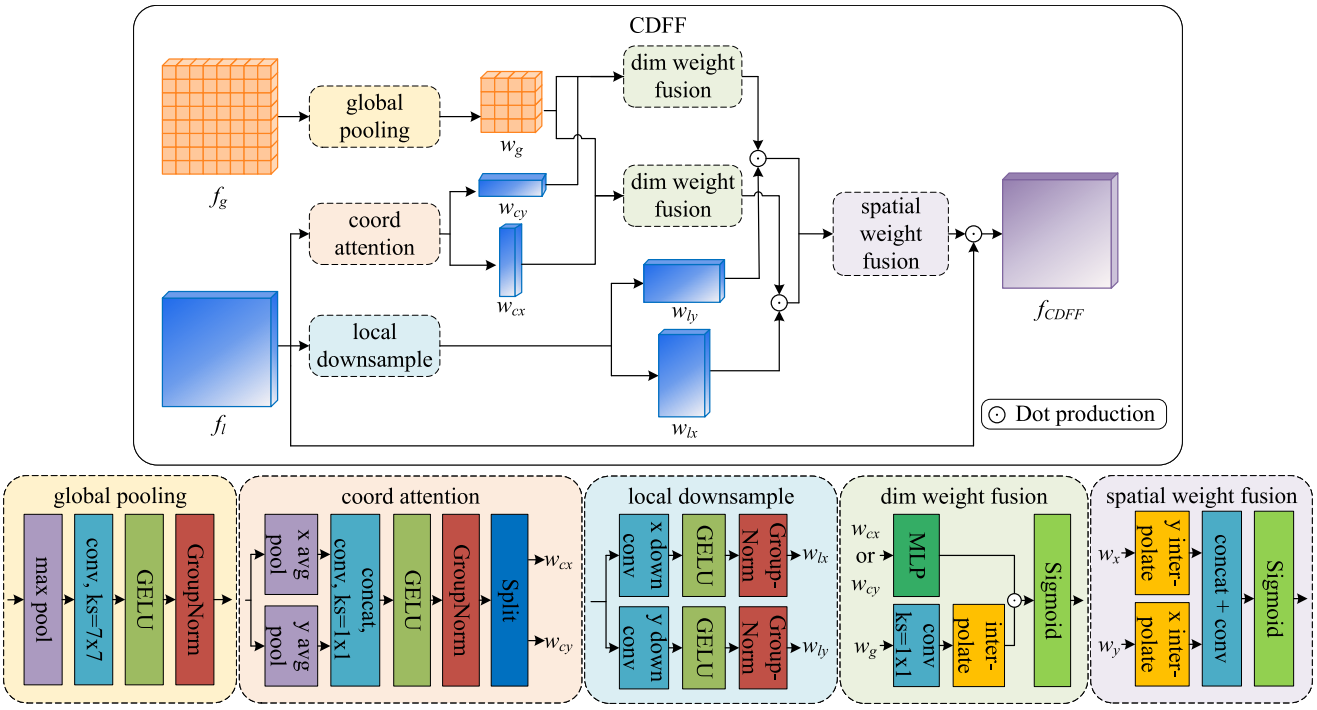


Fig. 2. Architecture of CDFF module. The GPB extracts weights from global features. The CAB and LDB are applied to generate local weights. Then, DWFB fuses the weights along different dimensions. Finally, SWFB integrates all weights, and the feature map is reweighted by the final weight map.

employment in the global branch. Comparatively, Mamba-based models provide a decent ability for global modeling with lower computational burden. Therefore, VMamba-t is chosen as the global branch. In the VMamba-t branch, the input image is first split into patches and then fed into several layers containing downsampling operation and a series of visual state-space blocks. The first four levels of both results from ResNet50 and VMamba-t are retained.

Specifically, given image $I \in \mathbb{R}^{3\times H_0\times W_0}$, where H_0 and W_0 are the height and width of the image, respectively, CNN branch and VMamba branch are used to extract local features $f_i^{\mathrm{l}} \in \mathbb{R}^{C_i^{\mathrm{l}}\times(H_0/2^{2+i})\times(W_0/2^{2+i})}$ and global features $f_i^{\mathrm{g}} \in \mathbb{R}^{C_i^{\mathrm{g}}\times(H_0/2^{2+i})\times(W_0/2^{2+i})}$, where $i \in \{0, 1, 2, 3\}$, $C_i^{\mathrm{l}} \in \{256, 512, 1024, 2048\}$, and $C_i^{\mathrm{g}} \in \{96, 192, 384, 768\}$. Two FPNs are then applied on them to map the channel number to 256, denoted as f_i^{ml} and f_i^{mg}.

With f_i^{ml} and f_i^{mg}, the CDFF module gathers necessary content and forms the fused features, denoted as f_i^{fused}. Self-attention and cross-fusion mechanisms are designed in this module to improve performance. Proposals p are obtained from the fused features f_i^{fused} by the RPN module. Then, RoIAlign operations with different resolutions extract RoI features from f_i^{ml}, f_i^{mg}, and f_i^{fused} according to p, to obtain the local, global, and fused RoI features, denoted as $\mathrm{roi}^{\mathrm{l}}$, $\mathrm{roi}^{\mathrm{g}}$, and $\mathrm{roi}^{\mathrm{fused}}$, respectively. In the refusion procedure for mask generation, the GFRM first gathers and enhances sensitive information through the interaction of $\mathrm{roi}^{\mathrm{l}}$, $\mathrm{roi}^{\mathrm{g}}$, and $\mathrm{roi}^{\mathrm{fused}}$, and the SCFF module finally integrates all the features to generate segmentation results.

In CDFF module, SCFF module, and GFRM, Transformer-based modules are applied to achieve feature fusion procedures. On the one hand, Transformer could be designed and implemented more flexibly. On the other hand, the following procedures after the backbone require relatively low computational cost, which is not affordable.

B. CDFF Module

After the backbone, the global and local features are obtained. To assemble more features for accurate detection, it is necessary to fuse the two sets of features effectively. However, simple operations such as dot production and a simple combination of concatenation and convolution are not adequate, which could lead to the loss of necessary features. Fusion procedures based on pooling operations and attention allow dynamic selection of important features and could be a more effective solution.

Inspired by Coordinate Attention [41], EMA [42], and TMAFNet [7], the CDFF module is proposed, which achieves feature fusion from the viewpoint of different dimensions based on dynamic selection. The structure of the CDFF module is illustrated in Fig. 2. Five key blocks are included in the module: global pooling block (GPB), coordinate attention block (CAB), local downsample block (LDB), dim weight fusion block (DWFB), and spatial weight fusion block (SWFB). The first three blocks are used to extract weights of different aspects from input features dynamically. Then, the last two blocks are used to mix the weights and generate the final weight map to reweight the input.

Specifically, given global features f^{g} and local features f^{l}, the global pooling block is first applied on f^{g} to generate global weight w_g. In this block, a max pooling of 2×2 is first used, and then, a convolution with a kernel size of 7×7 is used, followed by activation and normalization, which can be

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written as follows:

$$w_g = \mathrm{GN}\left(\phi\left(\mathrm{Conv}_{7\times7}\left(\mathrm{GMP}\left(f^{\mathrm{g}}\right)\right)\right)\right) \tag{1}$$

where GMP represents the global max pooling, ϕ represents the GELU, and GN represents the GroupNorm.

Then, the CAB and LDB are used to extract weights from local features f^{l}. The CAB first applies two pooling operations to the f^{l} along the X and Y dimensions. After that, these two results are concatenated and passed through the convolution layer, followed by the activation function and the normalization layer. The result is then split to generate two coordinate weights, $w_{\mathrm{cx}} \in \mathbb{R}^{C\times H}$ and $w_{\mathrm{cy}} \in \mathbb{R}^{C\times W}$. The procedure is shown as follows:

$$w_{\mathrm{cx}}, w_{\mathrm{cy}} = \mathrm{Split}\left(\mathrm{GN}\left(\phi\left(\mathrm{Conv}_{1\times1}\left(\mathrm{XAP}\left(f^{\mathrm{l}}\right)\copyright\mathrm{YAP}\left(f^{\mathrm{l}}\right)\right)\right)\right)\right) \tag{2}$$

where XAP and YAP represent the average pooling along X and Y dimensions, respectively. © represents the concatenation operation. Split represents that the feature map is split through the channel dimension.

After GPB and CAB, an LDB is applied to obtain weights with more detailed information. Instead of pooling operations, convolutions with kernels of different sizes are selected in this block to generate weight maps, which allows weights to focus more on relations of neighboring features and save more positional information.

Specifically, two convolution layers are applied to f^{l} to compress the feature vertically and horizontally, named Y-Down-Conv and X-Down-Conv. The kernel sizes of Y-Down-Conv and X-Down-Conv are 2×1 and 1×2, and the stride of each direction is the same as the kernel edge. Then, activation and normalization are used. The procedures are shown as follows:

$$w_{\mathrm{lx}} = \mathrm{GN}\left(\phi\left(\text{X-Down-Conv}\left(f^{\mathrm{l}}\right)\right)\right) \tag{3}$$

$$w_{\mathrm{ly}} = \mathrm{GN}\left(\phi\left(\text{Y-Down-Conv}\left(f^{\mathrm{l}}\right)\right)\right) \tag{4}$$

where w_{lx} and w_{ly} are the results of two procedures.

With weights of a different granularity, a fusion process is conducted to generate the final weight map. Two main steps are involved in this process, which are DWFB and SWFB. First, in DWFB, weights of the same dimension are fused with w_g. Then, in SWFB, the weights of two dimensions are fused to get the final weights. In the first step, interpolation is used to match w_g and dimensional weights on size. Then, pixelwise multiplication is conducted to finish the fusion. The procedures are as follows:

$$w_x = w_{\mathrm{lx}} \odot \sigma\left(\text{CW-Sum}\left(\text{X-UP}\left(\mathrm{Conv}_{1\times1}\left(w_g\right)\right) \odot w_{\mathrm{cx}}\right)\right) \tag{5}$$

$$w_y = w_{\mathrm{ly}} \odot \sigma\left(\text{CW-Sum}\left(\text{Y-UP}\left(\mathrm{Conv}_{1\times1}\left(w_g\right)\right) \odot w_{\mathrm{cy}}\right)\right) \tag{6}$$

where X-Up and Y-Up are the interpolations along X and Y dimensions, CW-Sum is the channelwise summation, $\odot$ represents the pixelwise multiplication, and σ represents the Sigmoid function. Before multiplication, w_{cx} and w_{cy} are broadcast to match the shapes.

In the second step, SWFB is applied to fuse weights of different dimensions. Two weight maps first align spatial shapes through interpolation. Then, the weights are concatenated and convolved followed by the activation function. Then, the dot product is applied to fuse the result and f^{l} to obtain the fusion result. The procedures are shown as follows:

$$w_{\mathrm{f}} = \sigma\left(\mathrm{Conv}_{3\times3}\left(\text{Y-UP}(w_x)\copyright\text{X-UP}\left(w_y\right)\right)\right) \tag{7}$$

$$f_{\mathrm{CDFF}} = f^{\mathrm{l}} \odot w_{\mathrm{f}} \tag{8}$$

where Y-Up and X-Up are the interpolation operations along the corresponding dimension, © is the concatenation operation, and f_{CDFF} is the final fusion result.

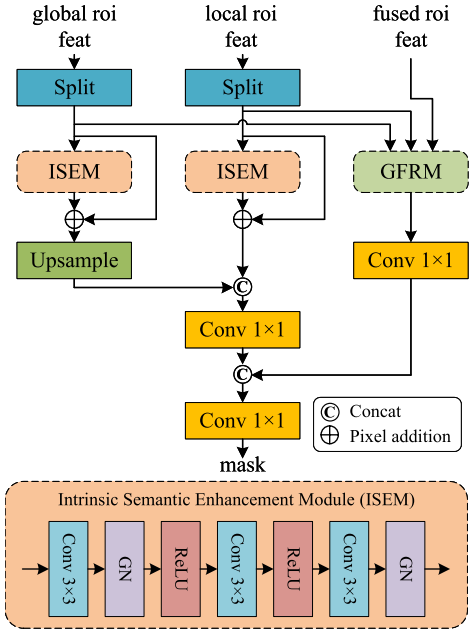


Fig. 3. Architecture of SCFF mask head.

C. SCFF Mask Head

For RSIs, the acquisition of instancewise accurate segmentation is challenging. Specifically, foreground and background parts are easy to confuse because of various shapes and complex background, which could affect the accuracy of boundary clarification. Synergistic usage of global and local features could improve the result, which could complement long-range features and local relationships of pixels. To efficiently leverage these features, the SCFF mask head is proposed to generate the final mask, as shown in Fig. 3. The features are split into two parts along channels. The first parts are processed by the intrinsic semantic enhancement module (ISEM) to augment semantic information. The second parts, together with the fused feature, are sent to the GFRM and calculate complementary semantic features. Then, a convolution layer is used to generate masks.

Specifically, after obtaining proposals, RoIAlign is applied to the three groups of feature maps according to the proposals. Three groups of RoI features with different resolutions are extracted, which are denoted as local RoI features r_l from local features, global RoI features r_g from global features, and fused RoI features r_{f} from fused features. Then, r_l and r_g are split into two features along the channel dimension, which are denoted as r_{ls}, r_{lc} and r_{gs}, r_{gc}, respectively. r_{ls} and r_{gs} are sent

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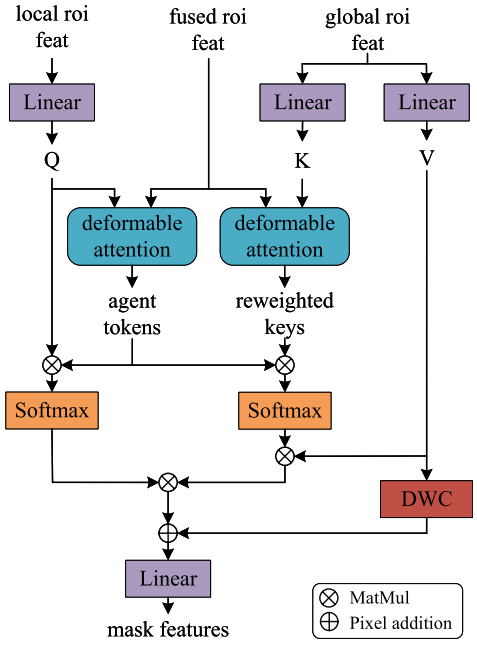


Fig. 4. Architecture of GFRM.

into ISEM for self-enhancement, and residual connections are applied to get the results f_{le} and f_{ge}. After matching sizes, the two outputs are concatenated and fused with convolution. On the other hand, r_{lc}, r_{gc}, and r_{f} are sent into GFRM to obtain fusion result f_{cross}. Finally, convolution is used on the concatenation results of f_{le}, f_{ge}, and f_{cross} to generate the mask.

GFRM is introduced in Section III-D. ISEM consists of convolutions, normalization, and activation functions, which are shown in Fig. 3.

D. Guided Feature Refusion Module

To blend meaningful features, the GFRM is proposed to efficiently fuse the complementary features, as shown in Fig. 4. In this module, agent attention [29] is applied to fuse two groups of features. Besides, deformable attention is incorporated into the module to improve the flexibility of feature assembly. The fused feature acts as guidance for selecting agent tokens from local features in place of the pooling operation. It is also used to reweight global features.

Specifically, r_{lc}, r_{gc}, and r_{f} are obtained as input. With linear layer, r_{lc} is transformed to query, and r_{gc} is transformed to key and value, which are denoted as Q, K, and V, which are shown as follows:

$$Q = r_{\text{lc}} W^q \tag{9}$$

$$K = r_{\text{gc}} W^k \tag{10}$$

$$V = r_{\text{gc}} W^v \tag{11}$$

where W^q, W^k, and W^v are the weights of linear layers.

Then, deformable attention modules are applied to select agent tokens and reweight key maps. Both modules take r_{f} as input and calculate cross-attention with Q and K, respectively, and generate agent tokens and reweighted keys, which are shown as follows:

$$A_t = \text{DeformAttn}(Q, r_{\text{f}}) \tag{12}$$

$$K_{\text{rw}} = \text{DeformAttn}(K, r_{\text{f}}) \tag{13}$$

where A_t represents the agent tokens and K_{rw} represents the reweighted keys.

After agent token sampling and key reweighting, two Softmax attention modules are used to generate agent features and cross features. Agent features are obtained from A_t, K_{rw}, and V with the first Softmax attention module. The attention of queries and agent tokens are calculated in the second module. Finally, cross features are generated based on fusion with agent bias and depthwise convolution, following the design of agent attention. The process can be expressed as follows:

$$A_{\text{f}} = \delta\left(\frac{A_t K_{\text{rw}}^T + B_1}{\sqrt{d}}\right) V \tag{14}$$

$$f_{\text{cross}} = \delta\left(\frac{\text{Q}A_t^T + B_2}{\sqrt{d}}\right) A_{\text{f}} + \text{DWC}(V) \tag{15}$$

where A_{f} is the agent features, δ represents the Softmax function, and d is the scaling factor.

E. Loss Function

The loss functions of this model are kept the same as the Cascade Mask R-CNN. To finish the training process, three constraints are applied to the model, which are the classification loss $\mathcal{L}_{\text{cls}}$, regression loss $\mathcal{L}_{\text{reg}}$, and segmentation loss $\mathcal{L}_{\text{segm}}$. $\mathcal{L}_{\text{cls}}$ is the cross-entropy loss between the predicted class and true labels. $\mathcal{L}_{\text{reg}}$ is the Smooth L1 loss between the predicted bounding boxes and ground truth. $\mathcal{L}_{\text{segm}}$ is also the cross-entropy loss between the predicted masks of each instance and real masks. The total loss is shown as follows:

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} + \lambda_{\text{segm}}\mathcal{L}_{\text{segm}} \tag{16}$$

where the hyperparameters λ_{cls}, λ_{reg}, and λ_{segm} control the tradeoff of the losses. In the experiments, all these parameters are set to be 1.

IV. Experiments

In this section, the performance of the proposed model is evaluated. Three datasets are used to evaluate our model and compare with current popular methods, which are iSAID dataset [30], NWPU VHR-10 dataset [31], and WHU Building dataset [33]. Besides, ablation experiments are conducted to validate the effect of the proposed modules.

A. Datasets

1) iSAID: An iSAID is a dataset based on DOTA [43] dataset. With classification and regression annotations of instance-level from the DOTA dataset, iSAID corrects the errors among them and complements the pixelwise masks for instance segmentation tasks. The dataset contains 2806 images of remote sensing. To save the computational cost of each image, the images are cropped into smaller patches of 800 × 800 with a stride of 200, and 655 451 instances are labeled on these patches. The patches are officially split into train,

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validation, and test sets. The training set contains 28 029 patches, validation contains 9512 patches, and the test set contains 19 377 patches. Because the annotations of the test set are not publicly available, the performances on the test set are reported from the official server, with inference results submitted and evaluated online.

2) NWPU VHR-10: NWPU VHR-10 is an RSI object detection dataset created by Northwestern Polytechnical University in 2014. The dataset consists of 715 optical RSIs from Google Earth and 85 pan-sharpened color infrared images from the Vaihingen dataset. Objects of ten classes are annotated in 650 images as positive samples, leaving the rest 150 images with only background as negative images. Mask labels are provided from [32]. In the experiments, the data are randomly divided into the training set and test set with a ratio of 8:2.

3) WHU Building: WHU Building dataset is a high-quality multisource dataset for building detection created by Wuhan University. In the experiment, the aerial image subset is applied, which contains 8188 optical images of 512×512 pixels. The images are captured from Christchurch, New Zealand, and contain various areas, such as countryside, residential, cultural, and industrial areas with resolutions from 0.0075 to 0.3 m. We keep the official split of the dataset into training, validation, and test sets. The training set contains 4736 images. The validation set contains 1036 images. The test set contains 2416 images. To generate instance-level annotations, OpenCV is used to transform the connected areas into instance annotations with the format of instance segmentation.

B. Evaluation Metrics

Following the evaluation metrics of COCO [48] dataset, we use average precision (AP) to evaluate the performances of models. AP is the average of the conditions where IoU thresholds range from 0.5 to 0.95 with a step size of 0.05. AP_{50} and AP_{75} represent the condition with IoU thresholds of 0.5 and 0.75. The results of bounding box regression and mask prediction are denoted as AP^{bbox} and AP^{mask}. To combine the above situations, six metrics are presented for each model, which is AP^{bbox}, AP^{bbox}_{50}, AP^{bbox}_{75}, AP^{mask}, AP^{mask}_{50}, and AP^{mask}_{75}.

C. Implementation Details

We choose Cascade Mask R-CNN as the baseline, and the number of bounding box heads is 3. When the number of mask heads is less than 3, the last one or two mask heads take the output features of corresponding bounding box heads as inputs. The model uses ResNet50 and VMamba-t as two branches of the backbone. Our model is built based on the MMDetection toolkit [49], and the experiments are conducted on a server with two RTX A6000 GPUs. We use AdamW [50] optimizer with an initial learning rate of 0.0001 and a weight decay of 0.005. We apply $1\times$ schedule as a training strategy for WHU Building datasets and the BITCC dataset, in which the model is trained for 12 epochs, and the learning rate is dropped by 0.1 in epochs 8 and 11. For NWPU VHR-10, we apply a $2\times$ schedule, where the number of total epochs is 24, and the learning rate is dropped by 0.1 in epochs 16 and 22.

D. Comparison With the State of the Art

In this section, we conduct a comprehensive quantitative analysis of the proposed model on three datasets and compare the results with mainstream methods. Both conventional methods for natural images and newly proposed task-specific methods for RSIs are included in the experiments. In conventional methods, we choose Mask R-CNN [44], Mask Scoring R-CNN [10], Cascade Mask R-CNN [11], HTC [12], CondInst [18], Mask2Former [51], and PatchDCT [13] as comparisons. Among the methods for RSI, we choose DB-BlendMask [25], ISDNet [20], CATNet [22], DCTC [9], Box2Mask [45], RSIISN [8], Liu et al. [40], FEA-Net [46], and OBBInst [47] as comparisons. Because some task-specific methods do not cover all datasets, and the computational resources are limited, some methods are omitted in the tables.

1) Results on iSAID Dataset: The results of our method and other SOTA methods on the iSAID dataset are shown in Table I. Beside the models for general image instance segmentation, DB-BlendMask [25], ISDNet [20], CATNet [22], DCTC [9], Box2Mask [45], RSIISN [8], FEA-Net [46], and OBBInst [47] are used for comparison. The RSIISN applies Mask R-CNN as the basic framework in the experiment. Two mask heads out of three are used in GLFRNet in the experiments to reduce computational cost.

As shown in Table I, the model achieves significant improvement of 6.9 on AP^{bbox} and 4.8 on AP^{mask} compared with Mask R-CNN. With the same basic architecture, the model surpasses Cascade Mask R-CNN by 4.5 on AP^{bbox} and 3.9 on AP^{mask}, and also surpasses HTC by 3.9 on AP^{bbox} and 3.6 on AP^{mask}. Considering models specified for RSI instance segmentation tasks, the model still obtains marginal improvements.

The results demonstrate that both global and local features are beneficial for performance improvement of instance segmentation. Our method of combining these two features could integrate the advantages of both and refine the results.

Compared with other methods, the proposed method could obtain better results on both the AP^{bbox}_{75} and AP^{mask}_{75}, which shows that the model performs better in accurate detection and segmentation.

2) Results on NWPU VHR-10 Dataset: The results of the methods on the NWPU VHR-10 dataset are shown in Table II. Besides the mainstream model for natural image instance segmentation, we use RSIISN, CATNet, DCTC, and FEA-Net as comparisons on this dataset. In this experiment, three segmentation heads are applied in the cascade form. The results with * show the proposed model with only one segmentation head, and the last line represents the model with three segmentation heads. The results show that the GLFRNet surpasses other models in both tasks of bounding box regression and mask segmentation, achieving 72.3 on AP^{bbox} and 69.6 on AP^{segm}, and surpasses the state-of-the-art (SOTA) method by 1.9 on bounding box regression and 4.1 on mask segmentation.

According to the results shown in Table II, it can be observed that our model achieves higher improvements on AP_{75} than AP_{50} in both tasks of regression and segmentation.

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TABLE I
OBJECT DETECTION AND INSTANCE SEGMENTATION ON iSAID DATASET

Method	Backbone	Set	AP^{bbox}	AP^{bbox}_{50}	AP^{bbox}_{75}	AP^{mask}	AP^{mask}_{50}	AP^{mask}_{75}
Mask R-CNN [44]	ResNet-50		40.9	61.5	45.7	35.8	58.2	38.1
Mask Scoring R-CNN [10]	ResNet-50		41.2	60.7	46.3	37.1	58.2	40.4
Cascade Mask R-CNN [11]	ResNet-50		43.3	62.5	48.0	36.7	58.8	39.8
HTC [12]	ResNet-50		43.7	63.0	48.5	37.0	59.4	39.8
Box2Mask-C [45]	ResNet-101	val	-	-	-	26.6	50.6	23.8
CondInst [18]	ResNet-50		37.5	59.0	40.5	30.6	54.3	30.0
PatchDCT [13]	ResNet-50		39.4	61.7	43.4	35.8	58.5	37.9
RSIISN [8]	Swin-T		45.0	**67.8**	51.5	39.2	64.0	43.3
DCTC [9]	DLASeg		-	-	-	36.3	58.2	38.0
GLFRNet(Ours)†	Dual Branch		**47.8**	67.3	**53.3**	**40.6**	**64.2**	**43.8**
DB-BlendMask [25]	ResNet-101		46.0	66.7	50.8	38.9	63.1	41.0
ISDNet [20]	ResNet-152		46.8	67.1	52.4	38.7	62.2	42.2
RSIISN [8]	Swin-T		45.9	**67.2**	52.3	40.3	**64.2**	43.4
CATNet [22]	ResNet-50	test	44.8	63.6	51.0	37.3	60.3	39.7
FEA-Net [46]	ResNet-101		-	-	-	38.7	61.5	41.2
OBBInst [47]	ResNet-101		-	-	-	23.9	45.5	24.9
GLFRNet(Ours)†	Dual Branch		**47.6**	66.3	**53.4**	**40.6**	63.1	**44.3**

† This model uses two mask heads in cascade form.

TABLE II
OBJECT DETECTION AND INSTANCE SEGMENTATION RESULTS ON NWPU VHR-10 TEST SET

Method	Backbone	AP^{bbox}	AP^{bbox}_{50}	AP^{bbox}_{75}	AP^{mask}	AP^{mask}_{50}	AP^{mask}_{75}
Mask R-CNN [44]	ResNet-50	65.8	90.4	78.9	63.7	90.2	71.0
Mask Scoring R-CNN [10]	ResNet-50	66.2	90.1	80.4	64.9	90.1	71.3
Cascade Mask R-CNN [11]	ResNet-50	68.5	89.5	81.6	63.2	89.2	69.3
HTC [12]	ResNet-50	69.4	91.0	83.1	64.2	91.0	70.0
CondInst [18]	ResNet-50	59.1	84.2	67.7	54.9	84.8	57.2
Mask2Former [51]	ResNet-50	61.3	78.9	70.7	62.2	84.3	72.3
PatchDCT [13]	ResNet-50	67.3	90.5	80.5	64.5	89.9	70.1
RSIISN [8]	Swin-T	64.1	91.1	75.4	65.5	91.3	73.1
CATNet [22]	ResNet-50	66.7	91.5	78.3	64.6	91.8	69.3
DCTC [9]	DLASeg	-	-	-	67.7	91.8	75.5
FEA-Net [46]	ResNet-101	-	-	-	64.3	92.8	70.3
GLFRNet(Ours)*	Dual Branch	71.0	93.6	83.6	67.7	92.9	74.9
GLFRNet(Ours)	Dual Branch	**72.3**	**93.9**	**86.3**	**69.6**	**94.0**	**81.3**

* This model only uses one mask head.

This phenomenon is also obvious when increasing the number of segmentation heads. This could prove the effectiveness of our model on accurate prediction in the instance segmentation task.

3) Results on WHU Building Dataset: The results of the methods on the WHU Building dataset are shown in Table III. Besides the mainstream methods, we apply RSIISN and models from Ye et al. [8] as RSI-specified methods for comparison. As shown in Table III, our model outperforms other methods, achieving 73.3 on bounding box regression and 70.6 in mask segmentation, outperforming the SOTA by 0.5 on object detection and 2.1 on segmentation.

As presented in Table III, the proposed method still shows the advantage of accurately performing the target tasks. Enhancements on AP^{bbox}_{75} and AP^{mask}_{75} demonstrate that the proposed method possesses a higher ability to accurately characterize the targets.

E. Qualitative Visual Comparison

In this section, we analyze and compare the results on the iSAID, NWPU VHR-10, and WHU Building datasets from different models in the visualized form to demonstrate the effectiveness of our model.

1) iSAID Dataset: The results from experiments on the iSAID dataset are shown in Fig. 5. As shown in Fig. 5, it is challenging for mainstream models to conduct detection and segmentation accurately in complex scenes of RSIs, while the proposed model improves the performances in these situations. Fig. 5(a) and (b) shows that classification for densely distributed small objects could be difficult for mainstream models. As presented in Fig. 5(c) and (d), these models also suffer from low accuracy in detecting and segmenting objects with complex and special shapes. Similar problems happen when processing objects of large scales and complex spatial relationships, which are shown in Fig. 5(e) and (f). Comparatively, our model performs better in these scenes, which demonstrates superiority in this task.

2) NWPU VHR-10 Dataset: The results on the NWPU VHR-10 dataset are shown in Fig. 6. Similar to the iSAID, the images in Fig. 6(a) and (b) containing planes and those in Fig. 6(c) and (d) with baseball diamonds pose high requirements for accurate segmentation. Images in Fig. 6(e) and (f)

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TABLE III
OBJECT DETECTION AND INSTANCE SEGMENTATION ON TEST SET IN WHU BUILDING DATASET

Method	Backbone	AP^{bbox}	AP^{bbox}_{50}	AP^{bbox}_{75}	AP^{mask}	AP^{mask}_{50}	AP^{mask}_{75}
Mask R-CNN [44]	ResNet-50	68.7	87.6	78.4	62.8	87.6	74.3
Mask Scoring R-CNN [10]	ResNet-50	68.6	87.3	78.1	63.1	87.3	73.6
Cascade Mask R-CNN [11]	ResNet-50	70.9	88.4	79.6	63.2	87.7	74.4
HTC [12]	ResNet-50	71.4	89.0	80.1	63.6	88.3	74.5
CondInst [18]	ResNet-50	73.1	**89.7**	81.1	60.1	87.6	71.0
Mask2Former [51]	ResNet-50	57.7	82.0	66.3	64.0	83.7	73.6
PatchDCT [13]	ResNet-50	70.6	89.3	80.0	69.3	89.3	79.0
Liu and Chen [40]	AR101-BPFPN	-	-	-	67.6	**90.0**	78.8
CATNet [22]	ResNet-50	70.1	87.7	80.0	62.5	87.1	73.9
RSIISN [8]	Swin-T	71.0	88.4	80.7	68.0	88.4	78.1
GLFRNet(Ours)*	Dual Branch	72.8	88.7	80.9	68.5	88.8	78.7
GLFRNet(Ours)	Dual Branch	**73.3**	88.8	**81.8**	**70.6**	88.9	**80.9**

* This model only uses one mask head.

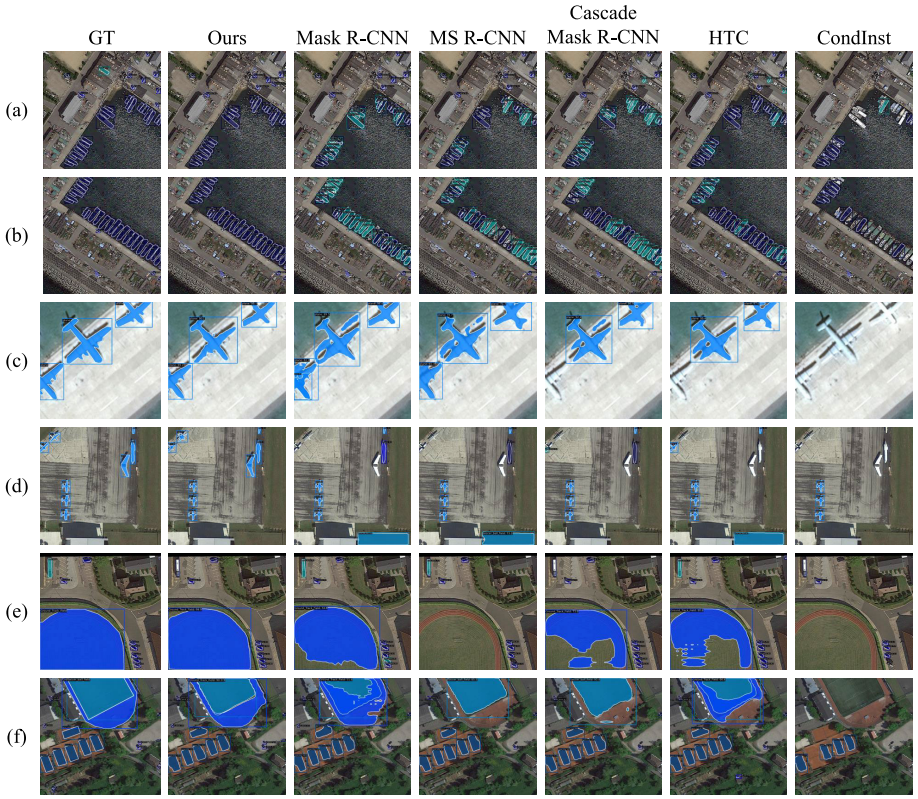


Fig. 5. Visualization results on iSAID dataset. (a) and (b) Superiority of classification for densely distributed small objects. (c) and (d) Superiority in detecting and segmenting objects with complex and special shapes. (e) and (f) Superiority processing objects of large scales and complex spatial relationships.

need identification ability for dense objects of the model. It can be observed in the images that our model surpasses the mainstream models in both detection and segmentation in these situations.

3) WHU Building Dataset: The results on the WHU Building dataset are shown in Fig. 7. As shown in the images, this dataset only contains one class of "Building," but the shapes of different instances vary greatly, which could be very challenging for models to generate masks of high quality. According to the results presented in Fig. 7(a) and (d), our model could distinguish similar contexts of foreground and background, and generate finer masks for the buildings. Images in Fig. 7(b) show the abilities of models to select correct pixels and render special shapes of buildings from complex contexts, and our model performs better than others. Fig. 7(c) and (f) demonstrates that our model has better detection accuracy than others on both aspects of true sample preservation and false sample suppression. Fig. 7(e) presents that our model has a better segmentation ability for dense objects. All these results could show that our model outperforms other ones.

F. Ablation Study

In this section, a series of experiments is conducted on the proposed model with the NWPU dataset to validate the significance of each module and parameters to the performance of the proposed method. In the experiments, the model uses Mask R-CNN as the basic framework. The training strategy of 2× with 24 training epochs is applied.

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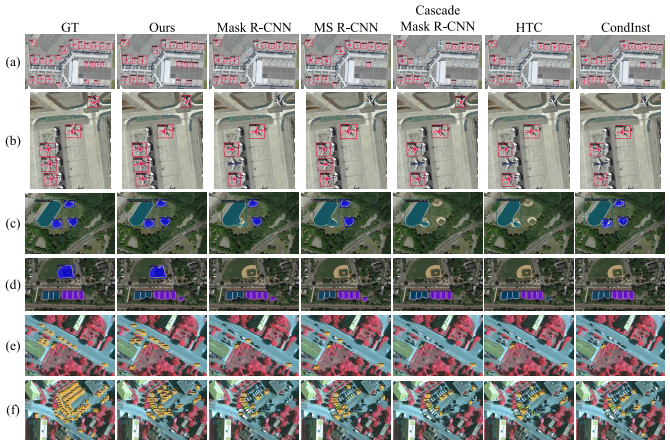


Fig. 6. Visualization results on NWPU VHR-10 dataset. (a) and (b) With planes and (c) and (d) with baseball diamonds show the superiority of accurate segmentation. (e) and (f) Superiority for identifying dense objects.



Fig. 7. Visualization results on WHU Building dataset. (a) and (d) Superiority for distinguishing similar contexts of foreground and background. (b) Ability to select correct pixels and render special shapes from complex contexts. (c) and (f) Better detection accuracy of true and false samples. (e) Better segmentation ability for dense objects.

TABLE IV
COMPARISON OF DIFFERENT BACKBONES

Backbone	AP^{bbox}	AP^{bbox}_{50}	AP^{bbox}_{75}	AP^{mask}	AP^{mask}_{50}	AP^{mask}_{75}
CNN	65.8	90.1	77.2	60.8	89.2	67.3
VMamba	69.2	93.1	82.8	66.7	92.9	73.5
dual branch	70.5	94.3	82.4	67.3	94.6	72.8

1) Effectiveness of Backbones: In the proposed model, two backbones are employed in the model to generate global and local features, which is based on the assumption that the application of comprehensive features could enhance the performance of the model. To validate the necessity of this architecture, we conduct experiments on models with the single backbone of ResNet50 and VMamba-t, and compare the results of the three methods. The results are shown in Table IV. The first row shows the result with only ResNet50 backbone, and the second row shows the results with only VMamba-t.

TABLE V
COMPARISON OF DIFFERENT FUSION METHODS FOR FPN

Fusion Method	AP^{bbox}	AP^{bbox}_{50}	AP^{bbox}_{75}	AP^{mask}	AP^{mask}_{50}	AP^{mask}_{75}
dot production	68.8	92.9	78.7	66.3	93.1	70.3
concat + conv	69.6	91.6	82.5	66.2	92.1	72.5
CDFF	70.5	94.3	82.4	67.3	94.6	72.8

TABLE VI
COMPARISON OF DIFFERENT FEATURES FOR MASK GENERATION

Features	AP^{bbox}	AP^{bbox}_{50}	AP^{bbox}_{75}	AP^{mask}	AP^{mask}_{50}	AP^{mask}_{75}
RoI features from FPN only	68.6	92.1	79.8	65.3	92.3	70.1
refusion features only	68.1	92.3	80.7	59.8	91.4	62.9
SCFF	70.5	94.3	82.4	67.3	94.6	72.8

TABLE VII
COMPARISON OF DIFFERENT FUSION METHODS FOR MASK GENERATION

Fusion Method	AP^{bbox}	AP^{bbox}_{50}	AP^{bbox}_{75}	AP^{mask}	AP^{mask}_{50}	AP^{mask}_{75}
concat + conv	68.8	91.9	80.6	65.3	91.9	69.8
dot mul + norm	70.2	93.2	81.6	66.7	92.7	71.7
add + conv	69.5	93.1	82.4	66.2	93.6	69.9
SCFF	70.5	94.3	82.4	67.3	94.6	72.8

It can be observed from the results that VMamba brings vast enhancement in regression and segmentation results compared to CNN, which shows the effectiveness of global features. However, it could also be seen that the model with dual branches performs best among these configurations, which demonstrates that both CNN and VMamba branches contribute to the results and could not be replaced by each other.

2) Methods for First Fusion Procedure: In the model, the CDFF module is designed to fuse the global and local features for the first time and generate features with complementary information for proposal generation. Besides this module, we also implement some simple ways for fusion, as shown in Table V. The first line represents that the fusion result is generated by dot production on the two feature maps of each level. The second line shows results from the feature map generated by the "concat then conv" procedure on the two features of each level.

Compared with the other results, our module surpasses these simple fusions in both bounding box prediction and mask segmentation. This shows that our module could preserve the important feature more efficiently with the designs of the attention mechanism.

3) Effectiveness of Different Features and Methods for Refusion Procedure: After proposals are obtained, RoI features are extracted from three groups of feature maps, and the SCFF module takes these RoI features as input to generate the prediction mask by self-enhancement and GFRM. However, the selection of applied feature maps and the design of interaction between them could affect the performance of mask segmentation. Hence, a series of experiments is conducted to verify the effectiveness of the proposed method on mask generation, which is shown in Tables VI and VII.

Table VI shows the performance of mask generation with different features input to the mask head. In the first row, the results are generated with extracted RoI features from global and local FPN outputs only, and the refusion procedure is not applied. The second row presents results generated solely

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TABLE VIII
PARAMETERS COUNT AND FPS ON NWPU VHR-10 DATASET

Method	Backbone	Params(M)	FPS
Mask R-CNN	ResNet-50	44.02	8.2
Mask Scoring R-CMM	ResNet-50	60.29	7.9
Cascade Mask R-CNN	ResNet-50	77.06	7.7
HTC	ResNet-50	77.19	8.0
CondInst	ResNet-50	**34.00**	10.5
Mask2Former	ResNet-50	44.01	6.3
CATNet	ResNet-50	54.66	4.9
RSIISN	Swin-T	53.67	6.8
DCTC	DLASet	51.25	**23.88**
GLFRNet(Ours)	Dual Branch	124.93	1.7

based on the refusion results. As presented in Table VI, using RoI features extracted from the FPN results directly could generate fair results. Comparatively, the mask result from refusion output only is much lower. The synergistic usage of the two types of features could obtain the best results, which is implemented by the SCFF module. These experiments demonstrate the necessity of complementing different features.

Table VII presents the results of several fusion methods to finish refusion. Results shown in the first row are acquired by the refusion procedure, which concats the fused, global, and local RoI features followed by convolutions. The second row shows the results from the refusion procedure of dot multiplication and normalization on the three RoI features. Similarly, the procedure with addition and convolution on the features generates results in the third row. Finally, results from the proposed refusion method are given. The results demonstrate that the specifically designed refusion procedure is more efficient on feature interaction and could generate more accurate results than other methods.

G. Quantitative Analysis of Computational Cost

We evaluate the parameter scale, FLOPs, and frames per second (FPS) in the inference of the proposed model and main-stream methods based on the NWPU VHR-10 dataset. The result is shown in Table VIII.

As the cost of improving performance, our model requires more resources to finish the inference procedure. The application of dual-branch backbone leads to a dramatic increase in the parameter scale. The complex architecture also results in the low speed in inference, which is shown in FPS.

V. CONCLUSION

This article proposes GLFRNet, an efficient instance segmentation model for RSIs of high accuracy. To address the challenges in RSIs, we propose the refusion process with three modules to fully leverage the global and local features from the dual-branch backbone. The CDFF module is used in the first fusion procedure for efficient fusion of global and local features from the two branches of the backbone. The SCFF module and GFRM form the refusion procedure to generate the masks of instances. The GFRM achieves a dynamic gathering of necessary detailed features from both global and local features guided by the fused features. The SCFF module finishes the mask generation by combining the global feature, local feature, and results from the GFRM. Extensive experiments demonstrate that GLFRNet surpasses the SOTA model by 1.9, 1.3, and 0.3 in mask APs on NWPU VHR-10, WHU Building, and iSAID datasets.

However, the model still has some issues to be improved. On the one hand, because of the dual-branch design in the backbone, the parameter number of the model is significantly larger than main-stream methods, which leads to higher requirements on computational resources and lower inference speed. On the other hand, many hand-craft models result in the complex implementation of the model. In the future, a more efficient model with automatic feature fusion is still to be explored.

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